Random Walk on Lattices and Graphs

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1 Introduction

A random walk describes the motion of a particle that moves step by step, choosing its direction at random.

This report aims to present our analysis of mathematical models of random walks on simple networks. We study random walks on the one-dimensional integer lattice $\mathbb{Z}$ (a straight line of points), the two-dimensional square lattice $\mathbb{Z}^2$ (a grid of points) and finite k-regular graphs, in which every node has exactly k neighbors and the walker moves from its current node to one of its neighboring nodes, chosen uniformly at random at each time step.

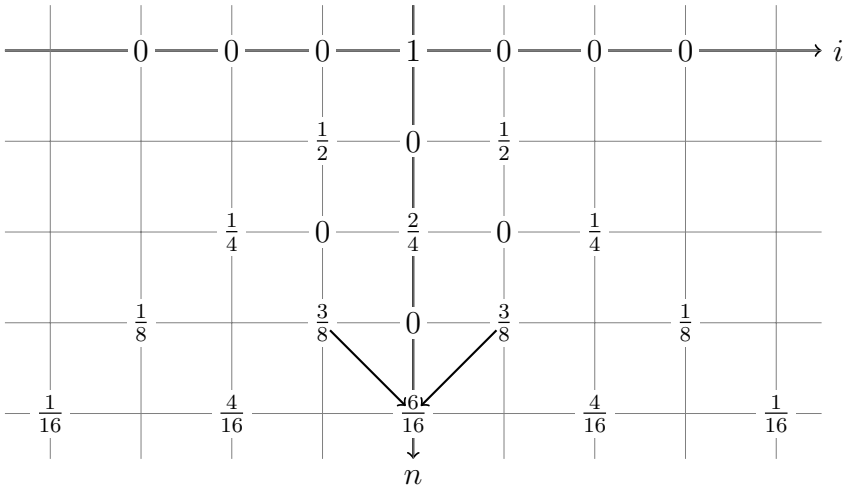
We will present our investigation on how far from the starting point the walker is likely to be after n steps, whether the walker eventually returns to the starting point, how long it takes on average to visit all nodes of a finite graph as the position of the walker changes over time, followed by some interesting scenarios which more closely resemble real life.

2 Random Walk on $\mathbb{Z}$

We define $p_n(i)$ as the probability that the walker is at node i after n steps. Hence, from the definition of a random walk on $\mathbb{Z}$, it follows that

$$p_n(i) = \frac{1}{2}p_{n-1}(i-1) + \frac{1}{2}p_{n-1}(i+1)$$

We know our initial condition yields $p_0(0) = 1$ and $\forall i \neq 0, p_0(i) = 0$. Our recurrence relation allows us to construct the following grid where each node represents the probability of finding the walker at the origin at that specific time, ignoring all previous moves.



Notice that the probabilities resemble the coefficients of a Pascal's triangle, with the denominator as powers of 2, so we can derive $p_n(i)$.
Specifically, suppose we took k right steps and $n-k$ left steps.
So $i = k-(n-k) = 2k-n$, or $k = \frac{n+i}{2}$

$$p_n(i) = \binom{n}{\frac{n+i}{2}} \left(\frac{1}{2}\right)^n$$

This applies when $n+i$ is even and $|i| \leq n$; otherwise, $p_n(i) = 0$.
Consider the expectation of each individual step.
$\mathbb{E}[X_i] = \frac{1}{2} - \frac{1}{2} = 0$.

$$\begin{aligned} \therefore \mathbb{E}[X_n] &= \mathbb{E}\left[X_0 + \sum X_i\right] \\ &= \mathbb{E}[X_0] + \sum 0 = \mathbb{E}[X_0] \quad \text{(by linearity of expectations)} \end{aligned}$$

Consider the variance of each individual step.
$\mathrm{Var}(X_i) = \mathbb{E}(X_i^2) - \mathbb{E}(X_i)^2 = 1 - 0 = 1$
So $\mathrm{Var}[X_n] = n$. Hence $\mathrm{Var}(X_n)$ grows linearly with n.
Consider $p_{2n}(0)$. Using the expression above, we get $p_{2n}(0) = \binom{2n}{n}\left(\frac{1}{2}\right)^{2n}$.
Stirling's Approximation: $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$

$$\begin{aligned} p_{2n}(0) &= \frac{(2n)!}{(n!)^2 2^{2n}} \\ &\sim \frac{\sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n}}{\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right)^2 2^{2n}} \\ &= \frac{\sqrt{4\pi n}\frac{2^{2n}n^{2n}}{e^{2n}}}{2\pi n\frac{n^{2n}}{e^{2n}} \cdot 2^{2n}} \\ &= \frac{1}{\sqrt{\pi n}} \end{aligned}$$

Notice that $\frac{1}{\sqrt{\pi n}}$ is a p-series where: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ $\begin{cases} \text{converges when } p > 1 \\ \text{diverges when } p \leq 1 \end{cases}$

Therefore, the series $\sum_{n=1}^{\infty} p_{2n}(0)$ diverges by comparison, which implies that the total expected number of returns is infinite. This occurs if and ony if a 1D random walk (on $\mathbb{Z}$) is **recurrent**.

3 Random Walk on $\mathbb{Z}^2$

For the proof, we use diagonal coordinates in which each move changes both components by ± 1; the precise rotation and scaling back to the standard axial walk are given below.

At each time step, the walker makes one diagonal move, changing both coordinates independently by either $+1$ or -1. Thus,

$$(\Delta X_k, \Delta Y_k) \in \{(1,1), (1,-1), (-1,1), (-1,-1)\},$$

with each outcome having probability $\frac{1}{4}$. Although both coordinates change, each diagonal move is counted as one time step.

Therefore, we can consider the set up as two independent 1D random walks occurring simultaneously.

$$q_n(i,j) = \frac{1}{4}\big(q_{n-1}(i-1,j-1) + q_{n-1}(i-1,j+1) + q_{n-1}(i+1,j-1) + q_{n-1}(i+1,j+1)\big)$$

Since

$$\mathbb{E}[\Delta X_k] = \mathbb{E}[\Delta Y_k] = \frac{1}{4}(1+1-1-1) = 0,$$

it follows that

$$\mathbb{E}[(X_n, Y_n)] = (0,0).$$

We then find the variance of the X-coordinate after 1 step.

$$\mathbb{E}[\Delta X_n^2] = 1^2 \cdot \frac{1}{2} + (-1)^2 \cdot \frac{1}{2} = 1$$
$$\hookrightarrow \mathbb{E}[X_n^2] = n$$

Similarly,

$$\mathbb{E}[Y_n^2] = n,$$

This means that

$$\mathbb{E}[X_n^2 + Y_n^2] = \mathbb{E}[X_n^2] + \mathbb{E}[Y_n^2]$$
$$= n + n = 2n.$$

Thus, as in one dimension, the mean-square displacement grows linearly with n.

Since X_n and Y_n are independent,

$$q_{2n}(0,0) = \binom{2n}{n}^2 \left(\frac{1}{4}\right)^{2n} = \left[\binom{2n}{n}\left(\frac{1}{2}\right)^{2n}\right]^2 = p_{2n}(0)^2$$

$$\hookrightarrow q_{2n}(0,0) \sim \left(\frac{1}{\sqrt{\pi n}}\right)^2 = \boxed{\frac{1}{\pi n}}$$

Therefore, $\sum_{n=1}^{\infty} q_{2n}(0,0)$ diverges by comparison with the harmonic series $\sum_{n=1}^{\infty} \frac{1}{\pi n}$. This proves that the 2D Random walk is always **recurrent**.

To make the change of convention explicit, let (X_n, Y_n) denote the coordinates used in the diagonal proof and let (U_n, V_n) denote the standard axial ("WASD") coordinates. Apply the linear transformation:

$$\begin{pmatrix} U_n \\ V_n \end{pmatrix} = A \begin{pmatrix} X_n \\ Y_n \end{pmatrix}, \qquad A = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Thus,

$$(1,1) \mapsto (1,0), \quad (1,-1) \mapsto (0,1), \quad (-1,1) \mapsto (0,-1), \quad (-1,-1) \mapsto (-1,0),$$

so the four diagonal moves become precisely the four standard axial moves. Moreover,

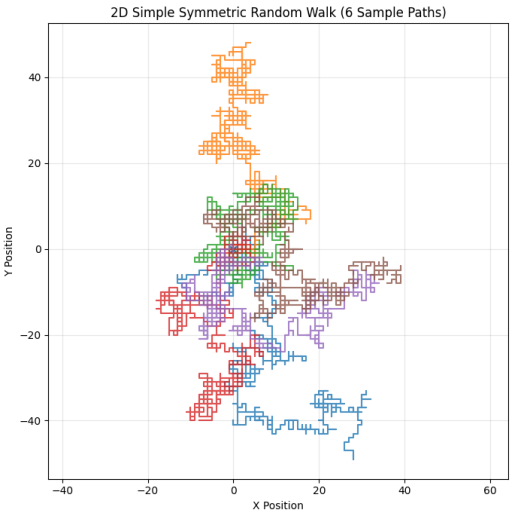
$$A^{\mathsf{T}}A = \frac{1}{2}I, \qquad U_n^2 + V_n^2 = \frac{1}{2}\left(X_n^2 + Y_n^2\right).$$

Hence the mean-square displacement is scaled from

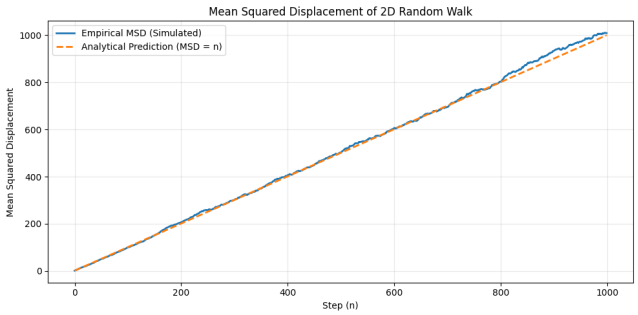
$$\mathbb{E}[X_n^2 + Y_n^2] = 2n \qquad \text{to} \qquad \mathbb{E}[U_n^2 + V_n^2] = \frac{1}{2}(2n) = n.$$

Since A is invertible and maps the origin to the origin, this change of coordinates does not affect the recurrence argument.

The simulations below use the standard axial ("WASD") walk with steps $(1,0)$, $(-1,0)$, $(0,1)$, and $(0,-1)$, rather than the diagonal convention used for the proof above. On the left are six sample paths. On the right, the mean-squared displacement averaged over 500 walks is plotted against the theoretical value $\mathbb{E}[U_n^2 + V_n^2] = n$. The two track each other closely, confirming that displacement grows linearly in n.



(a) Six sample paths



(b) Empirical and analytical mean-squared displacement

Figure 1: Numerical simulation of the conventional axial two-dimensional simple symmetric random walk.

3.1 Comparison of the 1D and 2D Walks

Property	**1D walk**	**2D diagonal walk**
Possible step	$\Delta X_k \in \{-1, 1\}$	$(\Delta X_k, \Delta Y_k) \in \{(\pm 1, \pm 1)\}$
Expected position	$\mathbb{E}[X_n] = 0$	$\mathbb{E}[(X_n, Y_n)] = (0, 0)$
Mean-square displacement	$\mathbb{E}[X_n^2] = n$	$\mathbb{E}[X_n^2 + Y_n^2] = 2n$
Return probability	$p_{2n}(0) \sim \dfrac{1}{\sqrt{\pi n}}$	$p_{2n}(0,0) \sim \dfrac{1}{\pi n}$
Return-probability series	$\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$	$\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$
Return-probability series	$\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$	$\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$

Table 1: Comparison of the one- and two-dimensional random walks.

Conclusion. Both return-probability series diverge; consequently, both walks are recurrent.

3.2 Fourier Transform Solutions for General Lattice Walks

Imagine we now have a symmetric random walk on $\mathbb{Z}^d$. Its movement probabilities satisfy

$$p_{n+1}(\mathbf{x}) = \frac{1}{2d} \sum_{j=1}^{d} \left[p_n(\mathbf{x} - \mathbf{e}_j) + p_n(\mathbf{x} + \mathbf{e}_j) \right], \tag{1}$$

where $\mathbf{e}_j$ denotes the unit vector in the j-th coordinate direction.

Multiplying both sides by $e^{i\mathbf{k}\cdot\mathbf{x}}$ and summing over all $\mathbf{x} \in \mathbb{Z}^d$ gives

$$\sum_{\mathbf{x}\in\mathbb{Z}^d} p_{n+1}(\mathbf{x}) e^{i\mathbf{k}\cdot\mathbf{x}} = \frac{1}{2d} \sum_{j=1}^{d} \sum_{\mathbf{x}\in\mathbb{Z}^d} \left[p_n(\mathbf{x} - \mathbf{e}_j) + p_n(\mathbf{x} + \mathbf{e}_j) \right] e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{2}$$

Define the characteristic function

$$\phi_n(\mathbf{k}) = \sum_{\mathbf{x}\in\mathbb{Z}^d} p_n(\mathbf{x}) e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{3}$$

$$\vec{x} \to \vec{x} - \vec{e}_j,$$

$$\sum_{\mathbf{x}\in\mathbb{Z}^d} p_n(\mathbf{x} - \mathbf{e}_j) e^{i\mathbf{k}\cdot\mathbf{x}} = e^{i\mathbf{k}\cdot\mathbf{e}_j} \phi_n(\mathbf{k}), \tag{4}$$

Similarly,

$$\sum_{\mathbf{x}\in\mathbb{Z}^d} p_n(\mathbf{x} + \mathbf{e}_j) e^{i\mathbf{k}\cdot\mathbf{x}} = e^{-i\mathbf{k}\cdot\mathbf{e}_j} \phi_n(\mathbf{k}). \tag{5}$$

Hence,

$$\phi_{n+1}(\mathbf{k}) = \frac{1}{2d} \sum_{j=1}^{d} \left(e^{i\mathbf{k}\cdot\mathbf{e}_j} + e^{-i\mathbf{k}\cdot\mathbf{e}_j} \right) \phi_n(\mathbf{k}) \tag{6}$$

$$= \frac{1}{d} \sum_{j=1}^{d} \cos(\mathbf{k}\cdot\mathbf{e}_j)\, \phi_n(\mathbf{k}). \tag{7}$$

Let

$$\mathbf{k}\cdot\mathbf{e}_j = k_j, \tag{8}$$

we obtain

$$\phi_{n+1}(\mathbf{k}) = \left(\frac{1}{d} \sum_{j=1}^{d} \cos k_j \right) \phi_n(\mathbf{k}). \tag{9}$$

If the walk begins at the origin, then

$$\phi_0(\mathbf{k}) = 1.$$

Therefore,

$$\boxed{\phi_n(\mathbf{k}) = \left(\frac{1}{d} \sum_{j=1}^{d} \cos k_j \right)^n}. \tag{10}$$

Applying the inverse Fourier transform gives

$$\boxed{p_n(\mathbf{x}) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \left(\frac{1}{d} \sum_{j=1}^{d} \cos k_j \right)^n e^{-i\mathbf{k}\cdot\mathbf{x}}\, d^d\mathbf{k}}. \tag{11}$$

This characteristic-function and Fourier-inversion approach follows the standard treatment of random walks on $\mathbb{Z}^d$; see Lawler and Limic [1, Section 2.2].

4 Random Walk on k-Regular Graphs

A finite k-regular graph is composed by N nodes labeled node 0, node 1, ..., and node $N-1$, where each node is connected to exactly k other nodes. Therefore, the transition matrix P of the graph is defined by

$$P_{ij} = \begin{cases} \frac{1}{k}, & \text{if nodes } i \text{ and } j \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases}$$

where $0 \le i, j \le N-1$

Let X_n be the position of the walker after n moves from the origin. Since the position of the walker is solely depend on the last position of the walker,

$$\Pr(X_{n+1} = j \mid X_n = i, X_{n-1}, \ldots, X_0) = \Pr(X_{n+1} = j \mid X_n = i).$$

Therefore, this process of the random walk is defined as a type of the Markov chain. Since the probabilities of every nodes in the graph is independent to n. This process is a time-homogeneous Markov chain.
In this research, two examples of k-regular random walk are considered: circle graph and complete graph.

4.1 Cycle Graph

Consider the cycle graph C_N with nodes

$$Nodes = \{0, 1, \ldots, N-1\},$$

where each node i is connected to nodes $i-1$ and $i+1$ modulus N, so C_N is a 2-regular graph.
At each step, the walker moves clockwise or anticlockwise with equal probability:

$$P_{ij} = \begin{cases} \frac{1}{2}, & j = i-1 \pmod N, \\ \frac{1}{2}, & j = i+1 \pmod N, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, the transition matrix is

$$P = \begin{pmatrix} 0 & \frac{1}{2} & 0 & \cdots & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & \cdots & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \frac{1}{2} & \vdots \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \cdots & \frac{1}{2} & 0 \end{pmatrix}.$$

The stationary distribution is uniform. Let

$$\pi = \left(\frac{1}{N}, \frac{1}{N}, \ldots, \frac{1}{N}\right).$$

For every vertex j,

$$(\pi P)_j = \frac{1}{N} \cdot \frac{1}{2} + \frac{1}{N} \cdot \frac{1}{2} = \frac{1}{N} = \pi_j.$$

Therefore,

$$\boxed{\pi_i = \frac{1}{N} \quad \text{for all } i.}$$

If there is another stationary distribution other than

$$\boxed{\pi_i = \frac{1}{N} \quad \text{for all } i.}$$

let $\pi_{i-1} < \pi_i$ let

$$d = \pi_i - \pi_{i-1}$$

since

$$\pi_i = \frac{\pi_{i-1} + \pi_{i+1}}{2}$$

$$\pi_{i+1} = \pi_i + d$$

Then,

$$\pi_{i+N} = \pi_i + Ni$$

but

$$\pi_{i+N} = \pi_i$$

Contradiction. Therefore,

$$\boxed{\pi_i = \frac{1}{N} \quad \text{for all } i.}$$

is the only stationary distribution of these k-regular random walk.

4.2 Complete Graph

In complete graph, every pair of nodes of the graph is connected. Therefore, the graph is $N-1$-regular.

The transition matrix is

$$P_{ij} = \begin{cases} \frac{1}{N-1}, & i \neq j, \\ 0, & i = j. \end{cases}$$

Equivalently,

$$P = \frac{1}{N-1}(J - I),$$

where J is the $N \times N$ matrix whose entries are all equal to 1.

Consider the stationary distribution of the graph. If the graph is stationary, then, π_i never changes.

$$\pi_i = \frac{\sum_{j \neq i} \pi_j}{N-1}$$
$$\frac{N}{N-1}\pi_i = \frac{\sum \pi_j}{N-1}$$
$$\frac{N}{N-1}\pi_i = \frac{1}{N-1}$$
$$\pi_i = \frac{1}{N}$$

Then, the stationary distribution of the graph is

$$\boxed{\pi = \left(\frac{1}{N}, \frac{1}{N}, \ldots, \frac{1}{N}\right).}$$

Start with node 0 with possibility 1. Define $P_k(i)$ be the possibility that the walker is on node i after k steps. Then,

$$P_0(0) = 1$$
$$P_0(i) = 0 \text{ if } i \neq 0$$

Then, $P_k(i)$ is symmetric when $i \neq 0$. Therefore,

$$P_{k+1}(0) = P_k(i), \ i \neq 0$$

When $i \neq 0$

$$P_k(i) = \frac{1 - P_k(0)}{N-1}$$

Therefore,

$$P_{k+1}(0) = \frac{1 - P_k(0)}{N-1}$$

When $i \neq 0$

$$P_{k+1}(i) = \frac{1 - P_{k+1}(0)}{N-1}$$

$$P_{k+1}(i) = \frac{1 - P_k(i)}{N-1}$$

Then, for all i,

$$P_{k+1}(i) = \frac{1 - P_k(i)}{N-1}$$

let

$$d_k(i) = |P_k(i) - \frac{1}{N}|$$

$$P_k(i) = \frac{1}{N} \pm d_k(i)$$

$$P_{k+1}(i) = \frac{1 - (\frac{1}{N} \pm d_k(i))}{N-1}$$

$$P_{k+1}(i) = \frac{(N-1) \pm N d_k(i))}{N(N-1)}$$

$$P_{k+1}(i) = \frac{1}{N} \pm \frac{d_k(i)}{N-1}$$

Therefore,

$$d_{k+1}(i) = \frac{d_k(i)}{N-1}$$

Then, the total difference after k steps $d_k = \sum d_k(i)$ follows

$$d_{k+1} = \frac{d_k}{N-1}$$

$$d_0 = (1 - \frac{1}{N}) + (N-1)(\frac{1}{N} - 0)$$

$$d_0 = 2\frac{N-1}{N}$$

$$d_k = 2\frac{N-1}{N} * (N-1)^{-k}$$

$$d_k = 2\frac{(N-1)^{1-k}}{N}$$

4.3 General Solution of the Difference Equation for a K-Regular Graph

$$p_{n+1} = p_n P$$

Therefore,

$$p_n = p_0 P^n.$$

For a k-regular undirected graph,

$$P = P^T.$$

Hence, P is a real symmetric matrix, so there exists an orthonormal basis of eigenvectors $v_1, v_2, \ldots, v_N$ such that

$$Pv_i = \lambda_i v_i,$$

and

$$v_i^T v_j = \begin{cases} 0, & i \neq j, \\ 1, & i = j. \end{cases}$$

The eigenvalues can be ordered as

$$1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq -1.$$

Since $v_1, \ldots, v_N$ form an orthonormal basis, the initial distribution can be expanded as

$$p_0 = \sum_{i=1}^{N} c_i v_i^T,$$

where

$$c_i = p_0 v_i.$$

Using the spectral decomposition,

$$P = Q \Lambda Q^T,$$

where

$$Q = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_N \\ | & | & & | \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N).$$

Therefore,

$$P^n = Q \Lambda^n Q^T.$$

Thus,

$$p_n = p_0 P^n = \sum_{i=1}^{N} (p_0 v_i) \lambda_i^n v_i^T.$$

Since

$$c_i = p_0 v_i,$$

the probability of being at vertex j after n steps is

$$\boxed{p_n(j) = \sum_{i=1}^{N} c_i \lambda_i^n v_i(j)}.$$

5 Variations in Random Walks

Consider a directed random walk on $\mathbb{Z}^2$ where, at each step, the walker moves in the positive x-direction with probability p, or the positive y-direction with probability $1-p$.

5.1 Directed Random Walk on $\mathbb{Z}^2$

Let S_n denote the state space of final positions after n steps. Because the walker only moves in positive directions, S_n is strictly bounded to the diagonal border $x + y = n$. For example, after 3 steps, the state space consists of exactly $N(S_3) = 4$ states: $S_3 = \{(3,0),(2,1),(1,2),(0,3)\}$.

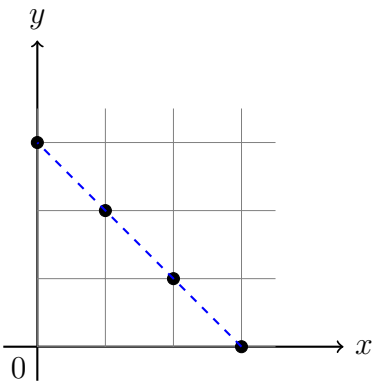


Figure 2: All valid possible final positions after 3 moves illustrated on a state space for a 2D random walk.

Generalizing to n steps, the total number of possible final states is $N(S_n) = n + 1$. Since the y-coordinate is only determined by the x-coordinate ($y = n - x$), the joint probability distribution of the final position simplies to a binomial distribution:

$$P(X = x \cap Y = y) = \binom{n}{x} p^x (1-p)^{n-x} \tag{12}$$

Let $Z = XY$ be the product of the final coordinates of the walker. Since the total steps $n = X + Y$ are conserved, X and Y are roots of the quadratic equation $t^2 - nt + Z = 0$. Applying the quadratic formula yields:

$$X, Y = \frac{n \pm \sqrt{n^2 - 4Z}}{2}$$

To find the distribution of Z, we sum the probabilities of the two distinct coordinate pairs that produce a given Z:

$$P(Z=z) = \binom{n}{\frac{n+\sqrt{n^2-4z}}{2}} p^{\frac{n+\sqrt{n^2-4z}}{2}}(1-p)^{\frac{n-\sqrt{n^2-4z}}{2}} + \binom{n}{\frac{n-\sqrt{n^2-4z}}{2}} p^{\frac{n-\sqrt{n^2-4z}}{2}}(1-p)^{\frac{n+\sqrt{n^2-4z}}{2}}$$

Because a step in the x-direction prohibits a step in the y-direction, X and Y are negatively correlated. We quantify this by calculating the covariance, $\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$.The independent expectations for each axis are simply $E[X] = np$ and $E[Y] = n(1-p)$. To find $E[XY]$, let $X_k, Y_k \in \{0,1\}$ be indicator variables for the k-th step in the x and y directions, respectively. Expanding the expectation of the product, we obtain:

$$E[XY] = E\left[\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{j=1}^{n} Y_j\right)\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} E[X_iY_j]$$

We partition the sum into simultaneous and distinct steps:

- **For** $i = j$: The walker cannot move in both directions in a single step, so $X_iY_i = 0$.
- **For** $i \neq j$: Distinct steps are independent, so $E[X_iY_j] = E[X_i]E[Y_j] = p(1-p)$. There are $n(n-1)$ such pairs.

Summing these cases yields $E[XY] = n(n-1)p(1-p)$.

$$\therefore \text{Cov}(X,Y) = E[XY] - E[X]E[Y] = n(n-1)p(1-p) - np \cdot n(1-p) = -np(1-p)$$

Alternatively, note that $X \sim B(n,p)$ as the number of trials is fixed and independent, and each step is a binary choice. Therefore, from fundamental properties of covariance, we obtain $\text{Cov}(X,Y) = \text{Cov}(X, n-X) = \text{Cov}(X,n) - \text{Cov}(X,X) = -Var(X) = -np(1-p)$.

5.2 Independent Walkers and Relative Distance

Consider two independent random walkers taking n steps. Let X_1 and X_2 denote their final x-coordinates, assuming step probabilities p_1 and p_2 in the positive direction. We want to find the distribution of their relative distance, $X = X_1 - X_2$. Conditioning on the second walker's position k, the distribution of X is:

$$P(\{X=x\}) = \sum_{k=1}^{n-x} P(X_1 = x+k)P(X_2 = k) = \sum_{k} \binom{n}{x+k} p_1^{x+k}(1-p_1)^{n-k-x} \cdot \binom{n}{k} p_2^{k}(1-p_2)^{n-k}$$

Setting $p_1 = p_2 = \frac{1}{2}$ simplifies this to:

$$P(\{X=x\}) = 2^{-2n} \sum_{k} \binom{n}{k}\binom{n}{x+k}$$

Theorem (Vandermonde's Identity)**.** *For any non-negative integers a, b, and c, the sum of the products of binomial coefficients is given by:*

$$\sum_{k=0}^{c} \binom{a}{k}\binom{b}{c-k} = \binom{a+b}{c}$$

(Vandermonde, 1772)

Proof. We will introduce the proof via a story. Consider $n = a + b$ committee members from two groups, each with a and b members respectively, from which we choose c members. Clearly, there are $\binom{a+b}{c}$ ways of doing this. Alternatively, we can consider selecting k people from the first group and the rest from the second, where k ranges through all possible values (as a summation), which completes the proof. □

Therefore, we can evaluate this sum using Vandermonde's Identity. By applying the symmetry of binomial coefficients, we can rewrite the sum as:

$$\begin{aligned}\sum_k \binom{n}{k}\binom{n}{x+k} &= \sum_k \binom{n}{k}\binom{n}{n-x-k} \\ &= \binom{2n}{n-x} = \binom{2n}{n+x}\end{aligned}$$

Thus, $P(\{X = x\}) = 2^{-2n}\binom{2n}{n+x}$. Letting $m = 2n$, we can write this more compactly as $P(\{X = x\}) = 2^{-m}\binom{m}{n+x}$.

5.3 Correlated Walkers

However, it is important to note that objects in real-world systems are often affected by many factors. Now suppose the walkers are correlated. Specifically, assume the second walker moves orthogonally to the first at each step, and let $p_1 = p$ be the probability of a move in either positive direction. Since the orthogonal movement means only one walker changes their x-coordinate per time step, the change in their relative distance (ΔX), per step is:

$$\Delta X = \begin{cases} +1 & \text{with probability } p \\ -1 & \text{with probability } 1-p \end{cases}$$

This reduces the relative distance X to a standard one-dimensional random walk, $X = \sum_{i=1}^{n} D_i$ where $D_i \in \{-1, 1\}$. The probability of reaching position x follows the binomial distribution:

$$P(\{X = x\}) = \binom{n}{\frac{n+x}{2}} p^{\frac{n+x}{2}} (1-p)^{\frac{n-x}{2}}$$

(Note: This probability $P(\{X = x\}) = 0$ if n and x have different parities). By linearity of expectation, the expected distance is:

$$E[X] = E\left[\sum_{i=1}^{n} D_i\right] = n \cdot (1(p) + (-1)(1-p)) = n(2p-1)$$

Hence we can see that the expected distance grows linearly with n, scaled by the bias $2p - 1$.

5.4 Vicious Walkers Escape or Not?

Problem. *In the two-dimensional random walk, the probabilities for A and B to move along the positive x-axis and positive y-axis in the first quadrant are p and 1-p respectively. A moves first and B moves later. B cannot move to the position where A is. When $T = t$, both A and B have exited the NM area. AB starts from $(0, 0)$. We want to find $P(T = t)$.*

In order to solve this problem we need to introduce a famous Lemma (*Lindström–Gessel–Viennot Lemma*).

Theorem (Lindström–Gessel–Viennot Lemma)**.** *In a directed acyclic graph: You have k starting points $A_1, A_2, \cdots, A_k$. You have k end points $B_1, B_2, \cdots, B_k$.*
The number of k paths from A_i to B_i ($i \in \{1, 2, 3, \cdots, k\}$) such that these paths are non-intersecting equals the determinant of a $k \times k$ matrix where the (i, j)-entry M_{ij} of the matrix is the total number of paths from A_i to B_j.

$$\textit{Number of non-intersecting paths} = \det(M) = \det \begin{pmatrix} N(A_1 \to B_1) & N(A_1 \to B_2) & \cdots \\ N(A_2 \to B_1) & N(A_2 \to B_2) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Example Proof: when $k = 2$ (as in my question),

$$\begin{aligned} \text{Number of non-intersecting paths} &= \det \begin{pmatrix} N(A_1 \to B_1) & N(A_1 \to B_2) \\ N(A_2 \to B_1) & N(A_2 \to B_2) \end{pmatrix} \\ &= \underbrace{N(A_1 \to B_1)N(A_2 \to B_2)}_{\text{some intersect, some not}} - \underbrace{N(A_2 \to B_1)N(A_1 \to B_2)}_{\text{all paths must intersect}} \end{aligned}$$

We can create a one-to-one mapping between the intersecting paths in $N(A_1 \to B_1)N(A_2 \to B_2)$ and the number of paths in $N(A_2 \to B_1)N(A_1 \to B_2)$. For any intersecting paths:

$$A_1 \to I \to B_1, \quad A_2 \to I \to B_2 \iff A_1 \to I \to B_2, \quad A_2 \to I \to B_1$$

I attempted to prove the Lemma in the case of k, however, it requires advanced content beyond my level of study which I cannot meaningfully comprehend within the short duration, therefore it is omitted.

At $t = 1$, we can assume $A \to A_0(1, 0)$. Then B has to go to $B_0(0, 1)$. It is also true when they swap positions, so we need to multiply by 2 on $p(1 - p)$.

Assume when $T = t$, A reaches $A_n(X_A, Y_A)$ and B reaches $B_n(X_B, Y_B)$.

$$X_A + Y_A = t, \quad X_B + Y_B = t \quad (X_A \neq X_B)$$

Regardless of intersection:

$$\begin{aligned} N(A_0 \to A_n) &= \binom{t-1}{X_A - 1} \\ N(B_0 \to B_n) &= \binom{t-1}{X_B} \\ N(A_0 \to B_n) &= \binom{t-1}{X_B - 1} \\ N(B_0 \to A_n) &= \binom{t-1}{X_A} \end{aligned}$$

By LGV $\implies$ Number of non-intersecting paths = $\det(M)$

$$M = \begin{pmatrix} N(A_0 \to A_n) & N(A_0 \to B_n) \\ N(B_0 \to A_n) & N(B_0 \to B_n) \end{pmatrix} = \begin{pmatrix} \binom{t-1}{X_A-1} & \binom{t-1}{X_B-1} \\ \binom{t-1}{X_A} & \binom{t-1}{X_B} \end{pmatrix}$$

So,

$$\begin{aligned} \det(M) &= \binom{t-1}{X_A-1}\binom{t-1}{X_B} - \binom{t-1}{X_A}\binom{t-1}{X_B-1} \\ &= \frac{1}{t^2}\binom{t}{X_A}\binom{t}{X_B}\Big[X_A(t-X_B) - X_B(t-X_A)\Big] \\ &= \frac{1}{t^2}\binom{t}{X_A}\binom{t}{X_B}[X_A t - X_B t] \\ &= \frac{X_A - X_B}{t}\binom{t}{X_A}\binom{t}{X_B} \end{aligned}$$

When $X_A < X_B$: After $T = k$, A, B lay on $x + y = k$. At $k = 1, X_A = 1, X_B = 0$, A is on the right of B.

① $k = t, X_A < X_B$, A is on the left of B. Define $d(k) = X_A(k) - X_B(k)$ for $k = 1, 2, \cdots, t$. $\Delta d(k) = d(k+1) - d(k)$, so $\Delta d(k) \in \{-1, 0, 1\}$. Since $d(1) > 0$ and $d(t) < 0$, from the Discrete Intermediate Value Theorem we can deduce $d(k_i) = 0$ ($k_i \in \{2, 3, \cdots, t-1\}$). So, A must intersect with B somewhere. $\det(M) = 0$.

② $X_A > X_B \implies$ paths $= \frac{X_A - X_B}{t}\binom{t}{X_A}\binom{t}{X_B}$

The probability of one path of A from $(1, 0)$ to $(X_A, Y_A) = p^{X_A - 1}(1-p)^{Y_A}$. The probability of one path of B from $(0, 1)$ to $(X_B, Y_B) = p^{X_B}(1-p)^{Y_B - 1}$. So, the probability of one pair of $(A, B) = p^{X_A + X_B - 1}(1-p)^{Y_A + Y_B - 1} = p^{X_A + X_B - 1}(1-p)^{2t - (X_A + X_B) - 1}$.

Therefore, the probability to reach (A_n, B_n) without intersection:

$$= 2 \cdot p(1-p)\frac{X_A - X_B}{t}\binom{t}{X_A}\binom{t}{X_B} \cdot p^{X_A + X_B - 1}(1-p)^{2t - (X_A + X_B) - 1}$$

If A_n and B_n are points out of the area "$N \times M$": $X > N$ or $Y > M$. $X > N$ or $t - X > M \implies X < t - M$.

$$P(T = t) = 2 \sum_{\substack{X_A \in (N, +\infty) \\ \cup (-\infty, t-M)}} \sum_{\substack{X_B \in (N, +\infty) \\ \cup (-\infty, t-M) \\ X_A > X_B}} \frac{X_A - X_B}{t}\binom{t}{X_A}\binom{t}{X_B} p^{X_A + X_B}(1-p)^{2t - (X_A + X_B)}$$

A Commented Python Code

```python
import numpy as np
import matplotlib.pyplot as plt

# Fix the pseudorandom-number sequence. This makes the figures exactly
# reproducible: every run uses the same randomly generated directions.
np.random.seed(42)

# Each row will represent one independent random walk, and each column
# will represent one time step within that walk.
n_steps = 1000
n_simulations = 500

# Draw an integer from {0, 1, 2, 3} independently for every step of every
# simulation. The four values represent right, left, up, and down, each
# with probability 1/4. The resulting shape is (500, 1000).
directions = np.random.randint(4, size=(n_simulations, n_steps))

# Convert the direction labels into horizontal increments:
# direction 0 -> +1, direction 1 -> -1, directions 2 or 3 -> 0.
# In NumPy, each comparison produces a Boolean array; converting it to
# integers changes True to 1 and False to 0.
dx = (directions == 0).astype(int) - (directions == 1).astype(int)

# Convert the same labels into vertical increments:
# direction 2 -> +1, direction 3 -> -1, directions 0 or 1 -> 0.
# Exactly one of dx and dy is nonzero at each step, so dx**2 + dy**2 = 1.
dy = (directions == 2).astype(int) - (directions == 3).astype(int)

# Cumulative sums convert the increments into positions. For example,
# x[r, n-1] is X_n for simulation r because array indices begin at zero.
x = np.cumsum(dx, axis=1)
y = np.cumsum(dy, axis=1)

# Plot six sample paths. These are individual realisations rather than
# averages, so they display the irregular motion characteristic of a walk.
plt.figure(figsize=(7, 7))
for i in range(6):
    plt.plot(x[i], y[i], alpha=0.8, linewidth=1.5)

# Mark the common initial position. The cumulative arrays begin after the
# first step, so the origin is added separately to make the start explicit.
plt.scatter(0, 0, marker="x")
plt.title("2D Simple Symmetric Random Walk (6 Sample Paths)")
plt.xlabel("X Position")
plt.ylabel("Y Position")
plt.axis("equal") # Use the same geometric scale on both coordinate axes.
plt.grid(alpha=0.3)
plt.tight_layout()
paths_file = "output.png"
```

```
plt.savefig(paths_file, dpi=180, bbox_inches="tight")
plt.show()

# Compute X_n^2 + Y_n^2 for every walk and time. Taking the mean along
# axis 0 averages over the 500 simulations while retaining all 1000 times.
empirical_msd = np.mean(x**2 + y**2, axis=0)

# For this axial walk, each step has squared length one and zero mean.
# Therefore E[X_n^2 + Y_n^2] = n. The first stored position is at n = 1,
# so the analytical array runs from 1 through n_steps rather than from 0.
analytical_msd = np.arange(1, n_steps + 1)

# Compare the Monte Carlo estimate with the exact theoretical prediction.
plt.figure(figsize=(10, 5))
plt.plot(empirical_msd, label="Empirical MSD (Simulated)", linewidth=2)
plt.plot(
    analytical_msd,
    label="Analytical Prediction (MSD = n)",
    linestyle="--",
    linewidth=2,
)
plt.title("Mean Squared Displacement of 2D Random Walk")
plt.xlabel("Step (n)")
plt.ylabel("Mean Squared Displacement")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
msd_file = "output (1).png"
plt.savefig(msd_file, dpi=180, bbox_inches="tight")
plt.show()

# Returning the filenames is convenient in notebook environments because
# it displays where the two generated figures have been saved.
(paths_file, msd_file)
```

References

[1] Gregory F. Lawler and Vlada Limic. *Random Walk: A Modern Introduction*, volume 123 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2010.